

Package ‘RFDose’

October 7, 2026

Type Package

Title Deterministic radiofrequency electromagnetic field (RF-EMF) dose calculator

Version 2.0.0

Description Package for calculating RF-EMF doses for cohorts.

License MIT + file LICENSE

Encoding UTF-8

LazyData true

Roxygen list(markdown = TRUE)

Imports dplyr,

tidyr,

utils,

yaml

Suggests knitr,

rmarkdown,

testthat (>= 3.0.0),

tibble

VignetteBuilder knitr

Depends R (>= 3.5)

Config/testthat/edition 3

Config/roxygen2/version 8.1.0

Contents

act_pwr_props	3
calculate_emf_doses	3
check_boolean_not_na	4
check_character_not_na	5
check_country	5
check_duration	6
check_headp_num	6
check_numeric_not_na	7
check_proportions	7
check_tissue	8
check_urbanicity	8
cordless_dose	9

dect_msar	10
dect_pwr	11
dect_sar	11
example_data	12
farfield_dose	13
farfield_msar	14
farfield_pwr	15
farfield_sar	15
fill_missing_variables	16
gaming_dose	17
headphones_dose	18
headphones_earset_dose	19
headphones_phone_dose	20
hotspot_dose	21
laptop_dose	22
laptop_msar	23
laptop_pwr	24
laptop_sar	25
load_params	26
load_tissue_params	27
location_props	27
mobilecall_dose	28
mobiledata_dose	30
mpc_bt_earset_msar	31
mpc_bt_phone_msar	32
mpc_msar	33
mpc_pwr_data	34
mpc_pwr_native	35
mpc_pwr_wifi	36
mpc_sar_data	36
mpc_sar_native	37
mpc_sar_wifi	38
mpd_msar	39
mpd_pwr_data	41
mpd_pwr_wifi	42
mpd_sar_data	43
mpd_sar_wifi	44
other_dose_wrapper	44
smartwatch_dose	46
smartwatch_phone_dose	47
smartwatch_watch_dose	48
tablet_dose	49
tablet_msar	50
tablet_pwr	51
tablet_sar	52
total_dose	53
vr_dose	54
wifi_dose	55
wifi_msar	56
wifi_pwr	57
wifi_sar	57

act_pwr_props	<i>Get low/lowmed/medhigh/high activity power proportions</i>
---------------	---

Description

Converts activity durations to proportions.

Usage

```
act_pwr_props(low_dur, lowmed_dur, medhigh_dur, high_dur)
```

Arguments

low_dur	Duration (in seconds per day) of low output power activities
lowmed_dur	Duration (in seconds per day) of low-medium output power activities
medhigh_dur	Duration (in seconds per day) of medium-high output power activities
high_dur	Duration (in seconds per day) of high output power activities

Value

Named list with the use proportion of each activity.

calculate_emf_doses	<i>Calculate Total RF-EMF Doses for ALL samples</i>
---------------------	---

Description

Wrapper function that calculates RF-EMF doses for a data frame of observations. This function will be removed in the future.

Usage

```
calculate_emf_doses(
  data,
  tissue,
  params = NULL,
  default_value_file = NULL,
  sar_file = NULL
)
```

Arguments

data	A data frame. Must have the exact same columns as the provided example dataset. Columns must not be missing. Missing values will be replaced with default values.
tissue	Tissue for which to calculate dose (default: "brain" or "body")
params	(experimental) Path to a parameter file in .yaml format. Must match the structure of the internal parameter file. Optional; if not specified, calculations use default parameters. SAR values may be left out (see sar_file).

default_value_file	List of default values to replace missing values (optional).
sar_file	(experimental) Path to a .yaml file with SAR values, matching the structure of the internal SAR files (e.g. sar_etain.yaml, sar_goliat_average.yaml). Optional; its values replace those of params, including device parameters it sets with its SAR values (e.g. placement proportions). If not specified, the SAR values of params are used, or the default SAR values (GOLIAT, average of four phantoms) if params has none. For the ETAIN SAR values, use sar_file = system.file("extdata", "sar_etain.yaml", package = "RFDose").

Details

Additional details will follow.

Value

A data frame. Columns contain the calculated RF-EMF dose of each row in mJ/kg/day, for each exposure source and each tissue.

check_boolean_not_na	<i>Check if value is logical/Boolean and non-NA</i>
----------------------	---

Description

Check if value is logical/Boolean and non-NA

Usage

```
check_boolean_not_na(x)
```

Arguments

x	Single value to be checked: should be a logical/Boolean, atomic, and not NA to pass check
---	---

Value

Returns invisible TRUE if check passes. Side effect of stopping calculation if check fails.

`check_character_not_na`*Check if value is character and non-NA*

Description

Check if value is character and non-NA

Usage

```
check_character_not_na(x)
```

Arguments

<code>x</code>	Single value to be checked: should be character, atomic, and not NA to pass check
----------------	---

Value

Returns invisible TRUE if check passes. Side effect of stopping calculation if check fails.

`check_country`*Check if country is valid*

Description

Checks if country code matches a country for which measurement data is available. Use "Other" for any other country.

Usage

```
check_country(country)
```

Arguments

<code>country</code>	Country to check. Must be one of these: "AT" (Austria), "BE" (Belgium), "FR" (France), "HU" (Hungary), "IT" (Italy), "NL" (Netherlands), "PL" (Poland), "ES" (Spain), "CH" (Switzerland), "UK" (United Kingdom), or "Other" (for any other country; calculations use averaged values from countries with available measurements)
----------------------	--

Value

Returns invisible TRUE if all checks pass. Side effect of stopping calculations if any check fails.

check_duration	<i>Check input duration(s)</i>
----------------	--------------------------------

Description

Checks supplied input duration or vector of durations; needed to stop calculation or raise warning in case of irregular input values.

Usage

```
check_duration(duration)
```

Arguments

duration	single duration (numeric) or vector of durations
----------	--

Value

Returns invisible TRUE if all checks pass. Side effects of stopping or showing warning message if any check fails

check_headp_num	<i>Check if number of headphones (earbuds) is valid</i>
-----------------	---

Description

Checks if the number of headphones (earbuds) worn during mobile phone calls is either 0, 1, or 2.

Usage

```
check_headp_num(headp_num)
```

Arguments

headp_num	Number of headphones (earbuds) worn during mobile phone call.
-----------	---

Value

Returns invisible TRUE if all checks pass. Side effect of stopping calculations if any check fails.

check_numeric_not_na	<i>Check if value is numeric and non-NA</i>
----------------------	---

Description

Check if value is numeric and non-NA

Usage

```
check_numeric_not_na(x)
```

Arguments

x	Single value to be checked: should be numeric, atomic, and not NA to pass check
---	---

Value

Returns invisible TRUE if check passes. Side effect of stopping calculation if check fails.

check_proportions	<i>Check input proportion(s)</i>
-------------------	----------------------------------

Description

Checks supplied input proportion or vector of proportions; needed to stop calculation or raise warning in case of irregular input values.

Usage

```
check_proportions(proportions)
```

Arguments

proportions	Single proportion value (between 0 and 1) or vector with multiple proportions to check
-------------	---

Value

Returns invisible TRUE if all checks pass. Side effects of stopping or showing warning message if any check fails.

check_tissue	<i>Check if tissue is valid</i>
--------------	---------------------------------

Description

Check if tissue with this name can be found in YAML file, and if it has any nested parameters. Parameters themselves are NOT checked for validity.

Usage

```
check_tissue(tissue, device, params)
```

Arguments

tissue	Name of tissue to be checked (character, length 1)
device	Device for which to check tissue (character, length 1)
params	Parameter list, as returned by load_params()

Value

Returns invisible TRUE if all checks pass. Side effect of stopping calculations if any check fails.

check_urbanicity	<i>Check if input urbanicity is valid</i>
------------------	---

Description

Checks if a supplied urbanicity value is valid (must be "urban", "rural", or "suburban").

Usage

```
check_urbanicity(urbanicity)
```

Arguments

urbanicity	Single urbanicity value (must be one of these: "urban", "rural", "suburban")
------------	--

Value

Returns invisible TRUE if checks pass. Side effect of stopping calculation if any check fails.

cordless_dose	<i>Calculate Dose from Cordless Calls</i>
---------------	---

Description

Calculates RF-EMF dose from making calls with cordless phone (also known as DECT phone).

Usage

```
cordless_dose(tissue, duration, ear_prop, params = load_params())
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
duration	Duration of cordless calls in s/day.
ear_prop	Proportion of time cordless phone is held against ear during calls.
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The dose is calculated as

$$Dose_{DECT} = mSAR_{DECT} * duration_{DECT}$$

Where:

- $Dose_{DECT}$ is the dose from DECT/cordless phone calls
- $mSAR_{DECT}$ is the momentary SAR value in mJ/kg
- $duration_{DECT}$ is the duration of the calls in s/day

Value

Tissue-specific RF-EMF dose from DECT/cordless phone calls in mJ/kg/day

See Also

[dect_msar\(\)](#)

Examples

```
cordless_dose(
  tissue      = "brain",
  duration    = 600,
  ear_prop    = 0.9)
```

dect_msar

*Calculate mSAR from Cordless Calls***Description**

Calculates mSAR (momentary SAR) from making calls with cordless phone (also known as DECT phone).

Usage

```
dect_msar(tissue, ear_prop, params = load_params())
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
ear_prop	Proportion of time cordless phone is held against ear during calls.
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The mSAR is calculated as

$$mSAR_{DECT} = nSAR_{DECT} * outputpower_{DECT}$$

Where:

- $mSAR_{DECT}$ is the mSAR from DECT/cordless phone calls
- $nSAR_{DECT}$ is the nSAR (normalized specific absorption rate) in W/kg/W
- $outputpower_{DECT}$ is the duration of the calls in mW

Value

Tissue-specific mSAR from DECT/cordless phone calls in mW/kg

See Also

[dect_pwr\(\)](#), [dect_sar\(\)](#)

Examples

```
dect_msar(
  tissue    = "brain",
  ear_prop  = 0.9)
```

dect_pwr	<i>Calculate Output Power of DECT/Cordless Phone</i>
----------	--

Description

Calculates the output power of DECT/cordless phone during a call.

Usage

```
dect_pwr(params = load_params())
```

Arguments

params Parameter list (optional). If not specified, calculations use default parameters.

Value

DECT/cordless phone output power in mW

dect_sar	<i>Calculate nSAR from Cordless Calls</i>
----------	---

Description

Calculates nSAR (normalized specific absorption rate) from making calls with cordless phone (also known as DECT phone).

Usage

```
dect_sar(tissue, ear_prop, params = load_params())
```

Arguments

tissue Tissue for which to calculate the nSAR (default: "brain" or "body")
 ear_prop Proportion of time cordless phone is held against ear during calls.
 params Parameter list (optional). If not specified, calculations use default parameters.

Value

Tissue-specific nSAR from DECT/cordless phone calls in W/kg/W

Examples

```
dect_sar(
  tissue      = "brain",
  ear_prop    = 0.9)
```

example_data	<i>Example data for testing</i>
--------------	---------------------------------

Description

A dataset containing example user data (made up)

Usage

example_data

Format

A data frame

sample_id test
use_5g test
travel_time test
urbanicity test
country test
headp_ea_num test
mpc_duration test
mpc_ea_prop test
mpc_headp_prop test
mpd_wifi_prop_home test
mpd_wifi_prop_travel test
mpd_wifi_prop_work test
mpd_dur_low test
mpd_dur_lowtomed test
mpd_dur_medtohigh test
mpd_dur_high test
dect_duration test
dect_ea_prop test
lptp_dur_low test
lptp_dur_lowtomed test
lptp_dur_medtohigh test
lptp_dur_high test
tblt_dur_low test
tblt_dur_lowtomed test
tblt_dur_medtohigh test
tblt_dur_high test
hotspot_duration test
smartwatch_duration test
tracker_duration test
vr_duration test
headphone_duration test
gaming_duration test

Source

Generated for package testing.

farfield_dose	<i>Calculate RF-EMF Dose from Far-field Sources</i>
---------------	---

Description

Calculates the tissue-specific RF-EMF dose from far-field sources

Usage

```
farfield_dose(tissue, country, urbanicity, travel_time, params = load_params())
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
country	Country of residence (Austria = "AT", Belgium = "BE", France = "FR", Hungary = "HU", Italy = "IT", Netherlands = "NL", Poland = "PL", Spain = "ES", Switzerland = "CH", United Kingdom = "UK", elsewhere = "Other")
urbanicity	Urbanicity of home / workplace (rural, suburban, or urban)
travel_time	Time spent commuting in seconds per day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from far-field sources is calculated as:

$$Dose_{farfield} = mSAR_{farfield} * 86400$$

Where:

- $Dose_{farfield}$ is the dose from far-field sources
- $mSAR_{farfield}$ is the momentary SAR value in mJ/kg

Value

Tissue-specific RF-EMF dose from far-field sources in mJ/kg/day

See Also

[farfield_msar\(\)](#)

Examples

```
farfield_dose(
  tissue      = "body",
  country     = "CH",
  urbanicity  = "suburban",
  travel_time = 1800)
```

farfield_msar	<i>Calculate mSAR of Far-field Sources</i>
---------------	--

Description

Calculates the mSAR of far-field sources

Usage

```
farfield_msar(tissue, country, urbanicity, travel_time, params = load_params())
```

Arguments

tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
country	Country of residence (Austria = "AT", Belgium = "BE", France = "FR", Hungary = "HU", Italy = "IT", Netherlands = "NL", Poland = "PL", Spain = "ES", Switzerland = "CH", United Kingdom = "UK", elsewhere = "Other")
urbanicity	Urbanicity of home / workplace (rural, suburban, or urban)
travel_time	Time spent commuting in seconds per day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends on the country, urbanicity, and time spent commuting.

Value

mSAR in W/kg/W/m**2

See Also

[farfield_pwr\(\)](#), [farfield_sar\(\)](#)

Examples

```
farfield_msar(
  tissue      = "body",
  country     = "CH",
  urbanicity  = "suburban",
  travel_time = 1800)
```

farfield_pwr

Calculate Output Power of Far-field Sources

Description

Calculates the output power of far-field sources

Usage

```
farfield_pwr(country, urbanicity, travel_time, params = load_params())
```

Arguments

country	Country of residence (Austria = "AT", Belgium = "BE", France = "FR", Hungary = "HU", Italy = "IT", Netherlands = "NL", Poland = "PL", Spain = "ES", Switzerland = "CH", United Kingdom = "UK", elsewhere = "Other")
urbanicity	Urbanicity of home / workplace (rural, suburban, or urban)
travel_time	Time spent commuting in seconds per day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends on the country, urbanicity, and time spent commuting.

Value

Output power in mW/m**2

Examples

```
farfield_pwr(
  country      = "CH",
  urbanicity   = "suburban",
  travel_time  = 1800)
```

farfield_sar

Calculate nSAR from Far-Field Sources

Description

Calculates tissue-specific nSAR from far-field sources.

Usage

```
farfield_sar(tissue, params = load_params())
```

Arguments

tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue.

Value

nSAR in W/kg/W/m**2

Examples

```
farfield_sar(
  tissue      = "brain")
```

```
fill_missing_variables
```

Fill missing values in input data frame with missing values

Description

Replaces missing values in input data frame with values specified as default values. In case a column is completely missing, it is added to the data frame, with all values being the default value.

Usage

```
fill_missing_variables(data, defaults, warn_threshold = 0.1)
```

Arguments

data	Input data frame
defaults	Named list (key = column name, value = default value to replace missing values in this column)
warn_threshold	threshold proportion of missing data per column to raise warning

Value

data frame with missing data replaced with default values. Side effect of raising warning if proportion of missing data exceeds warn_threshold for at least one column.

gaming_dose	<i>Calculate RF-EMF Dose from online gaming (console)</i>
-------------	---

Description

Calculates the tissue-specific RF-EMF dose from online gaming

Usage

```
gaming_dose(tissue, duration_gaming, params = load_params())
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
duration_gaming	duration of gaming with portable console in s/day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from online gaming is calculated as:

$$Dose_{gaming} = mSAR_{gaming} * duration_{gaming}$$

Where:

- $mSAR_{gaming}$ is the mSAR of gaming
- $Dduration_{gaming}$ is the duration of gaming with portable console in s/day

Value

Tissue-specific RF-EMF dose from gaming in mJ/kg/day

Examples

```
gaming_dose(
  tissue      = "body",
  duration_gaming = 600)
```

headphones_dose	<i>Calculate RF-EMF Dose from Bluetooth Headphones</i>
-----------------	--

Description

Calculates the tissue-specific RF-EMF dose from bluetooth headphones (calls excluded)

Usage

```
headphones_dose(tissue, duration_headphones, params = load_params())
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
duration_headphones	duration of time Bluetooth headphones were used in s/day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from Bluetooth headphone use is calculated as:

$$Dose_{headphones} = Dose_{earset} + Dose_{phone}$$

Where:

- $Dose_{earset}$ is the dose contribution from the headphones themselves
- $Dose_{phone}$ is the dose contribution from the phone connection to the headphones

Value

Tissue-specific RF-EMF dose from Bluetooth headphone use in mJ/kg/day

See Also

[headphones_earset_dose\(\)](#), [headphones_phone_dose\(\)](#)

Examples

```
headphones_dose(
  tissue          = "body",
  duration_headphones = 6000)
```

headphones_earset_dose

Calculate RF-EMF Dose from Smart Watch (earset contribution only)

Description

Calculates the tissue-specific RF-EMF dose from Bluetooth headphone use (earset contribution only).

Usage

```
headphones_earset_dose(tissue, duration_headphones, params = load_params())
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
duration_headphones	duration of time Bluetooth headphones were worn in s/day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from Bluetooth headphone use (earset contribution) is calculated as:

$$Dose_{earset} = mSAR_{earset} * duration_{earset}$$

Where:

- $mSAR_{earset}$ is the momentary SAR for the headphone connection
- $duration_{earset}$ is the use duration in s/day

Value

Tissue-specific RF-EMF dose from Bluetooth headphone use (earset contribution only) in mJ/kg/day

Examples

```
headphones_earset_dose(
  tissue = "body",
  duration_headphones = 6000)
```

headphones_phone_dose *Calculate RF-EMF Dose from Smart Watch (phone contribution only)*

Description

Calculates the tissue-specific RF-EMF dose from Bluetooth headphone use (phone contribution only).

Usage

```
headphones_phone_dose(tissue, duration_headphones, params = load_params())
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
duration_headphones	duration of time Bluetooth headphones were worn in s/day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from Bluetooth headphone use (phone contribution) is calculated as:

$$Dose_{phone} = mSAR_{phone} * duration_{phone}$$

Where:

- $mSAR_{phone}$ is the momentary SAR for the headphone connection
- $duration_{phone}$ is the use duration in s/day

Value

Tissue-specific RF-EMF dose from Bluetooth headphone use (phone contribution only) in mJ/kg/day

Examples

```
headphones_phone_dose(
  tissue          = "body",
  duration_headphones = 6000)
```

hotspot_dose	<i>Calculate RF-EMF Dose from Mobile Phone Hotspot</i>
--------------	--

Description

Calculates the tissue-specific RF-EMF dose from hotspot

Usage

```
hotspot_dose(tissue, duration_hotspot, params = load_params())
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
duration_hotspot	duration of time hotspot was active in s/day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from hotspot use is calculated as:

$$Dose_{hotspot} = mSAR_{hotspot} * duration_{hotspot}$$

Where:

- $mSAR_{hotspot}$ is the mSAR of hotspot use
- $Dduration_{hotspot}$ is the duration of hotspot use in s/day

Value

Tissue-specific RF-EMF dose from hotspot use in mJ/kg/day

Examples

```
hotspot_dose(
  tissue      = "body",
  duration_hotspot = 600)
```

laptop_dose

*Calculate RF-EMF Dose from Laptop Use***Description**

Calculates the tissue-specific RF-EMF dose from laptop use.

Usage

```
laptop_dose(
  tissue,
  dur_low,
  dur_lowmed,
  dur_medhigh,
  dur_high,
  params = load_params()
)
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
dur_low	Duration (in seconds per day) of low output power activities on laptop
dur_lowmed	Duration (in seconds per day) of low-medium output power activities on laptop
dur_medhigh	Duration (in seconds per day) of medium-high output power activities on laptop
dur_high	Duration (in seconds per day) of high output power activities on laptop
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The laptop RF-EMF dose is calculated as:

$$Dose_{laptop} = mSAR_{laptop} * duration_{laptop}$$

Where:

- $Dose_{laptop}$ is the dose from laptop use
- $mSAR_{laptop}$ is the momentary SAR value in mJ/kg
- $duration_{laptop}$ is the duration of laptop use

We distinguish between 4 types of activities:

1. Low output power: sending e-mails, browsing the internet, scrolling and chatting on social media, sending text messages
2. Low to medium output power: online gaming, streaming music, sending voice messages
3. Medium to high output power: watching videos, uploading pictures or videos, making video calls
4. High output power: uploading large files

Value

Tissue-specific RF-EMF dose from laptop use in mJ/kg/day

See Also

[laptop_msar\(\)](#)

Examples

```
laptop_dose(
  tissue      = "brain",
  dur_low     = 0,
  dur_lowmed  = 1230,
  dur_medhigh = 0,
  dur_high    = 0)
```

laptop_msar

Calculate mSAR from Laptop Use

Description

Calculates the tissue-specific momentary SAR (mSAR) from laptop use.

Usage

```
laptop_msar(
  tissue,
  dur_low,
  dur_lowmed,
  dur_medhigh,
  dur_high,
  params = load_params()
)
```

Arguments

tissue	Tissue for which to calculate mSAR (default: "brain" or "body")
dur_low	Duration (in seconds per day) of low output power activities on laptop
dur_lowmed	Duration (in seconds per day) of low-medium output power activities on laptop
dur_medhigh	Duration (in seconds per day) of medium-high output power activities on laptop
dur_high	Duration (in seconds per day) of high output power activities on laptop
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The mSAR is calculated as:

$$mSAR_{laptop} = nSAR_{laptop} * outputpower_{laptop}$$

Where:

- $mSAR_{laptop}$ is the mSAR from laptop use
- $nSAR_{laptop}$ is the normalised SAR value in mJ/kg
- $outputpower_{laptop}$ is the duration of laptop use

We distinguish between 4 types of activities:

1. Low output power: sending e-mails, browsing the internet, scrolling and chatting on social media, sending text messages
2. Low to medium output power: online gaming, streaming music, sending voice messages
3. Medium to high output power: watching videos, uploading pictures or videos, making video calls
4. High output power: uploading large files

Value

Tissue-specific mSAR from laptop use in mW/kg

See Also

[laptop_pwr\(\)](#), [laptop_sar\(\)](#)

Examples

```
laptop_msar(
  tissue      = "brain",
  dur_low     = 0,
  dur_lowmed  = 1230,
  dur_medhigh = 0,
  dur_high    = 0)
```

laptop_pwr

Calculate Laptop Output Power

Description

Calculates the laptop output power during use

Usage

```
laptop_pwr(
  band,
  dur_low,
  dur_lowmed,
  dur_medhigh,
  dur_high,
  params = load_params()
)
```

Arguments

band	WiFi frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz)
dur_low	Duration (in seconds per day) of low output power activities on laptop
dur_lowmed	Duration (in seconds per day) of low-medium output power activities on laptop
dur_medhigh	Duration (in seconds per day) of medium-high output power activities on laptop
dur_high	Duration (in seconds per day) of high output power activities on laptop
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends on the frequency band (2.4 GHz or 5.0 GHz) and the type of activity.

Value

Output power in mW

Examples

```
laptop_pwr(
  band      = "2",
  dur_low   = 0,
  dur_lowmed = 1230,
  dur_medhigh = 0,
  dur_high  = 0)
```

laptop_sar

Calculate nSAR during Laptop Use

Description

Calculates tissue-specific nSAR from laptop use.

Usage

```
laptop_sar(tissue, band, params = load_params())
```

Arguments

tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
band	Frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz)
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue and the frequency band.

Value

nSAR in W/kg/W

Examples

```
laptop_sar(  
  tissue    = "brain",  
  band      = "2")
```

load_params	<i>Load dose calculation parameters</i>
-------------	---

Description

Loads default parameters or parameters from external source and returns them as a nested list.

Usage

```
load_params(path = NULL, sar_path = NULL)
```

Arguments

path	Path of parameter file to load (must be a YAML file). If null, default non-SAR parameters are loaded. The file may contain SAR values or not.
sar_path	Path of a YAML file with SAR values (optional), with the structure devices > device > tissue > SAR values. Its values replace those of the parameters loaded from path. Besides SAR values, it may set device parameters that go with them (e.g. placement proportions such as legs_prop).

Details

Default parameters are split in two files: non-SAR parameters (params.yaml) and SAR values, which are merged into a single list. The default SAR values are those of GOLIAT, averaged over four phantoms (sar_goliat_average.yaml). The ETAIN SAR values are in sar_etain.yaml.

SAR values are taken from sar_path if given. Otherwise, the SAR values of the parameter file are used, or the default SAR values if the parameter file has none. A SAR file may also set device parameters that go with its SAR values (e.g. placement proportions), which replace those of the parameter file.

Value

List of parameters loaded from yaml input file(s)

See Also

[load_tissue_params\(\)](#)

load_tissue_params	<i>Load tissue-specific parameters</i>
--------------------	--

Description

Loads device- and tissue-specific parameters (SAR values) and returns them as a list.

Usage

```
load_tissue_params(params, device_type, tissue_name)
```

Arguments

params	List of parameters (default parameter file or custom YAML that matches default's structure)
device_type	Name of device (default:)
tissue_name	Name of tissue (default: "brain" or "body")

Value

List of parameters (SAR values) specific to selected device type and tissue.

See Also

[load_params\(\)](#)

location_props	<i>Calculate proportion of time spent at home vs work vs outdoors vs travelling</i>
----------------	---

Description

Based on travel time (input variable by user). Assumes a fixed ratio of time spent at home vs at work vs outdoors.

Usage

```
location_props(travel_time, home_prop, work_prop, outd_prop)
```

Arguments

travel_time	daily time spent travelling / commuting in seconds (s)
home_prop	proportion of time spent at home WITHOUT considering commute
work_prop	proportion of time spent at work WITHOUT considering commute
outd_prop	proportion of time spent outside WITHOUT considering commute

Details

We assume that the time spent commuting takes away from time spent at home, but not the time spent at work or outdoors.

Value

List with following elements: "travel", "home", "work", "out", corresponding value is the proportion of time spent in each location.

mobilecall_dose	<i>Calculation of RF-EMF Dose from Mobile Phone Calls</i>
-----------------	---

Description

Includes RF-EMF exposure from phone itself, for Bluetooth connection to headphones, and from headphones themselves

Usage

```
mobilecall_dose(
  tissue,
  duration,
  ear_prop,
  headp_prop,
  urbanicity,
  use_5g,
  travel_time,
  headp_ear_num,
  wifi_prop_home,
  wifi_prop_work,
  wifi_prop_travel,
  params = load_params()
)
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
duration	Duration of mobile phone call in seconds per day
ear_prop	Proportion of call performed with phone held against ear
headp_prop	Proportion of call performed with active Bluetooth connection to headphones
urbanicity	Urbanicity of home / workplace (rural, suburban, or urban)
use_5g	TRUE if 5g services are used for calls, FALSE if not

travel_time	Time spent commuting in seconds per day
headp_ear_num	Number of Bluetooth headphones used during call
wifi_prop_home	Proportion of time connected to WiFi (vs mobile data) at home
wifi_prop_work	Proportion of time connected to WiFi (vs mobile data) at school / work
wifi_prop_travel	Proportion of time connected to WiFi (vs mobile data) while commuting
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The mobile call RF-EMF dose is calculated as:

$$Dose_{mobilecall} = Dose_{phone} + Dose_{bluetooth}$$

Where:

- $Dose_{mobilecall}$ is the total dose from all mobile call activities
- $Dose_{phone}$ is the dose from the mobile phone (no Bluetooth)
- $Dose_{bluetooth}$ is the dose from using Bluetooth headphones during the call.

For each source (phone and bluetooth), the dose is calculated as:

$$mSAR * duration$$

Where:

- $mSAR$ is the momentary SAR value in mJ/kg
- $duration$ is the call duration in seconds

Value

Tissue-specific RF-EMF dose in mJ/kg/day

See Also

[mpc_msar\(\)](#)

Examples

```
mobilecall_dose(
  tissue      = "brain",
  duration    = 400,
  ear_prop    = 0.67,
  headp_prop  = 0.17,
  urbanicity  = "suburban",
  use_5g      = TRUE,
  travel_time = 1800,
  headp_ear_num = 2,
  wifi_prop_home = 1,
  wifi_prop_work = 0.8,
  wifi_prop_travel = 0)
```

mobiledata_dose

*Calculation of RF-EMF Dose from Mobile Phone Data Use***Description**

The mobile data RF-EMF dose is calculated as:

$$Dose_{data} = mSAR_{data} * duration_{data}$$

Usage

```
mobiledata_dose(
  tissue,
  duration_low,
  duration_lowmed,
  duration_medhigh,
  duration_high,
  use_5g,
  wifi_prop_home,
  wifi_prop_work,
  wifi_prop_travel,
  urbanicity,
  travel_time,
  params = load_params()
)
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
duration_low	Duration (in seconds per day) of low output power activities
duration_lowmed	Duration (in seconds per day) of low-medium output power activities
duration_medhigh	Duration (in seconds per day) of medium-high output power activities
duration_high	Duration (in seconds per day) of high output power activities
use_5g	TRUE if 5G services are used for data transfer, FALSE if not
wifi_prop_home	Proportion of time connected to WiFi (vs mobile data) at home
wifi_prop_work	Proportion of time connected to WiFi (vs mobile data) at school / work
wifi_prop_travel	Proportion of time connected to WiFi (vs mobile data) while commuting
urbanicity	Urbanicity of home / workplace (rural, suburban, or urban)
travel_time	Time spent commuting in seconds per day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

Where:

- $Dose_{data}$ is the dose from mobile data use
- $mSAR_{data}$ is the momentary SAR value in mJ/kg
- $duration_{data}$ is the duration of mobile data use

We distinguish between 4 types of mobile data use:

1. Low output power: sending e-mails, browsing the internet, scrolling and chatting on social media, sending text messages
2. Low to medium output power: online gaming, streaming music, sending voice messages
3. Medium to high output power: watching videos, uploading pictures or videos, making video calls
4. High output power: uploading large files

Value

Tissue-specific RF-EMF dose from data use in mJ/kg/day

See Also

[mpd_msar\(\)](#)

Examples

```
mobiledata_dose(
  tissue      = "brain",
  duration_low    = 4860,
  duration_lowmed = 540,
  duration_medhigh = 4860,
  duration_high   = 540,
  use_5g         = FALSE,
  wifi_prop_home  = 0.5,
  wifi_prop_work  = 0.5,
  wifi_prop_travel = 0.5,
  urbanicity     = "suburban",
  travel_time     = 1800)
```

mpc_bt_earset_msar	<i>Calculate mobile call mSAR for Bluetooth headphones (headphones only)</i>
--------------------	--

Description

Calculates the mSAR for mobile calling using Bluetooth headphones (headphone/earset only; the contribution of the mobile phone is calculated separately).

Usage

```
mpc_bt_earset_msar(tissue, headp_ear_num, params = load_params())
```

Arguments

tissue	Tissue for which to calculate mSAR (default: "brain" or "body")
headp_ear_num	Number of Bluetooth earphones worn during call (0, 1, or 2)
params	Parameter list (optional). If not specified, calculations use default parameters.

Value

mSAR from Bluetooth calls (headphones/earset only) in mJ/kg/day)

See Also

[mpc_bt_phone_msar\(\)](#)

Examples

```
mpc_bt_earset_msar(
    tissue = "brain"
)
```

mpc_bt_phone_msar	<i>Calculate mobile call dose from Bluetooth headphones (phone only)</i>
-------------------	--

Description

Calculates the RF-EMF dose from mobile calling using Bluetooth headphones (phone contribution only; the contribution of the headphones/earset is calculated separately).

Usage

```
mpc_bt_phone_msar(tissue, headp_ear_num, params = load_params())
```

Arguments

tissue	Tissue for which to calculate mSAR (default: "brain" or "body")
headp_ear_num	Number of Bluetooth earphones worn during call (0, 1, or 2)
params	Parameter list (optional). If not specified, calculations use default parameters.

Value

Dose from Bluetooth calls (phone contribution only) in mW/kg)

See Also

[mpc_bt_earset_msar\(\)](#)

Examples

```
mpc_bt_phone_msar(
    tissue = "brain"
)
```

mpc_msar

*Calculation of mSAR from Mobile Phone Calls***Description**

Calculates mSAR (momentary specific absorption rate) from mobile calls for a specific tissue.

Usage

```
mpc_msar(
  tissue,
  prop_native,
  prop_data,
  prop_wifi,
  headp_prop,
  ear_prop,
  speaker_prop,
  urbanicity,
  use_5g,
  travel_time,
  params = load_params()
)
```

Arguments

tissue	Tissue for which to calculate mSAR (default: "brain" or "body")
prop_native	Proportion of mobile call using native connection
prop_data	Proportion of mobile call using mobile data connection
prop_wifi	Proportion of mobile call using WiFi connection
headp_prop	Proportion of call performed with active Bluetooth connection to headphones
ear_prop	Proportion of call performed with phone held against ear
speaker_prop	Proportion of call performed in speaker mode
urbanicity	Urbanicity of home / workplace (rural, suburban, or urban)
use_5g	TRUE if 5g services are used for calls, FALSE if not
travel_time	Time spent commuting in seconds per day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The mSAR is calculated as

$$mSAR = nSAR_{native} \times outputpower_{native} + nSAR_{data} \times outputpower_{data} + nSAR_{wifi} \times outputpower_{wifi}$$

where

- $nSAR_{native}$, $nSAR_{data}$, $nSAR_{wifi}$ are the nSAR (normalized specific absorption rates, in W/kg/W) from mobile phone calls using native, mobile data, or WiFi networks, respectively
- $outputpower_{native}$, $outputpower_{data}$, $outputpower_{wifi}$ are the output powers in mW of the device (mobile phone) using native, mobile data, or WiFi networks, respectively

Value

Momentary SAR (mSAR) in mW/kg

See Also

[mpc_pwr_native\(\)](#), [mpc_pwr_data\(\)](#), [mpc_pwr_wifi\(\)](#), [mpc_sar_native\(\)](#), [mpc_sar_data\(\)](#), [mpc_sar_wifi\(\)](#)

Examples

```
mpc_msar(  
  tissue      = "body",  
  prop_native = 0.7,  
  prop_data   = 0.2,  
  prop_wifi   = 0.1,  
  headp_prop  = 0.17,  
  ear_prop    = 0.67,  
  speaker_prop = 0.17,  
  urbanicity  = "suburban",  
  use_5g      = TRUE,  
  travel_time = 1800)
```

mpc_pwr_data	<i>Calculate Mobile Phone Output Power during Data Mobile Phone Calls</i>
--------------	---

Description

Calculates the mobile phone output power during calls using mobile data.

Usage

```
mpc_pwr_data(band, urbanicity, travel_time, params = load_params())
```

Arguments

- band Technology (3g, 4g, or 5g) used for the call
- urbanicity Urbanicity of home / workplace (rural, suburban, or urban)
- travel_time Time spent commuting in seconds per day
- params Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends in the technology used and the location in which the call is performed (urbanicity, indoors/outdoors/commuting)

Value

Output power in mW

Examples

```
mpc_pwr_data(  
  band      = "4g",  
  urbanicity = "suburban",  
  travel_time = 1800)
```

mpc_pwr_native	<i>Calculate Mobile Phone Output Power during Native Mobile Phone Calls</i>
----------------	---

Description

Calculates the mobile phone output power during calls using native network.

Usage

```
mpc_pwr_native(band, urbanicity, travel_time, params = load_params())
```

Arguments

band	Technology (2G, 3g, 4g, or 5g) used for the call
urbanicity	Urbanicity of home / workplace (rural, suburban, or urban)
travel_time	Time spent commuting in seconds per day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends in the technology used and the location in which the call is performed (urbanicity, indoors/outdoors/commuting)

Value

Output power in mW

Examples

```
mpc_pwr_native(  
  band      = "4g",  
  urbanicity = "suburban",  
  travel_time = 1800)
```

mpc_pwr_wifi	<i>Calculate Mobile Phone Output Power during WiFi Mobile Phone Calls</i>
--------------	---

Description

Calculates the mobile phone output power during calls using WiFi connection.

Usage

```
mpc_pwr_wifi(band, params = load_params())
```

Arguments

band	Frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz) used for the call
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends on the frequency band (2.4 GHz or 5.0 GHz)

Value

Output power in mW

Examples

```
mpc_pwr_wifi(
  band = "5")
```

mpc_sar_data	<i>Calculate nSAR during Data Mobile Phone Calls</i>
--------------	--

Description

Calculates tissue-specific nSAR from mobile phone calls using mobile data.

Usage

```
mpc_sar_data(
  tissue,
  band,
  headp_prop,
  ear_prop,
  speaker_prop,
  params = load_params()
)
```

Arguments

tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
band	Technology (2G, 3g, 4g, or 5g) used for the call
headp_prop	Proportion of call performed with active Bluetooth connection to headphones
ear_prop	Proportion of call performed with phone held against ear
speaker_prop	Proportion of call performed in speaker mode
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue, the technology (2G, 3g, 4g or 5g), and the location of the phone during the call (against ear, with Bluetooth headphones, in speaker mode)

Value

nSAR in W/kg/W

Examples

```
mpc_sar_data(
  tissue      = "brain",
  band        = "4g",
  headp_prop  = 0.17,
  ear_prop    = 0.67,
  speaker_prop = 0.17)
```

mpc_sar_native

Calculate nSAR during Native Mobile Phone Calls

Description

Calculates tissue-specific nSAR from mobile phone calls using native network.

Usage

```
mpc_sar_native(
  tissue,
  band,
  headp_prop,
  ear_prop,
  speaker_prop,
  params = load_params()
)
```

Arguments

tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
band	Technology (2G, 3g, 4g, or 5g) used for the call
headp_prop	Proportion of call performed with active Bluetooth connection to headphones
ear_prop	Proportion of call performed with phone held against ear
speaker_prop	Proportion of call performed in speaker mode
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue, the technology (2G, 3g, 4g or 5g), and the location of the phone during the call (against ear, with Bluetooth headphones, in speaker mode)

Value

nSAR in W/kg/W

Examples

```
mpc_sar_native(  
  tissue      = "brain",  
  band        = "4g",  
  headp_prop  = 0.17,  
  ear_prop    = 0.67,  
  speaker_prop = 0.17)
```

mpc_sar_wifi	<i>Calculate nSAR during WiFi Mobile Phone Calls</i>
--------------	--

Description

Calculates tissue-specific nSAR from mobile phone calls using WiFi connection.

Usage

```
mpc_sar_wifi(  
  tissue,  
  band,  
  headp_prop,  
  ear_prop,  
  speaker_prop,  
  params = load_params()  
)
```

Arguments

tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
band	Frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz) used for the call
headp_prop	Proportion of call performed with active Bluetooth connection to headphones
ear_prop	Proportion of call performed with phone held against ear
speaker_prop	Proportion of call performed in speaker mode
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue, the technology (2.4GHz or 5.0GHz), and the location of the phone during the call (against ear, with Bluetooth headphones, in speaker mode)

Value

nSAR in W/kg/W

Examples

```
mpc_sar_wifi(
  tissue      = "brain",
  band        = "2",
  headp_prop  = 0.17,
  ear_prop    = 0.67,
  speaker_prop = 0.17)
```

mpd_msar

Calculation of mSAR from Mobile Phone Data

Description

Calculates mSAR (momentary specific absorption rate) from data (mobile data and WiFi) use for a specific tissue

Usage

```
mpd_msar(
  tissue,
  duration_low,
  duration_lowmed,
  duration_medhigh,
  duration_high,
  use_5g,
  wifi_prop_home,
  wifi_prop_work,
  wifi_prop_travel,
  urbanicity,
  travel_time,
  params = load_params()
)
```

Arguments

tissue	Tissue for which to calculate mSAR (default: "brain" or "body")
duration_low	Duration (in seconds per day) of low output power activities
duration_lowmed	Duration (in seconds per day) of low-medium output power activities
duration_medhigh	Duration (in seconds per day) of medium-high output power activities
duration_high	Duration (in seconds per day) of high output power activities
use_5g	TRUE if 5G services are used for data transfer, FALSE if not
wifi_prop_home	Proportion of time connected to WiFi (vs mobile data) at home
wifi_prop_work	Proportion of time connected to WiFi (vs mobile data) at school / work
wifi_prop_travel	Proportion of time connected to WiFi (vs mobile data) while commuting
urbanicity	Urbanicity of home / workplace (rural, suburban, or urban)
travel_time	Time spent commuting in seconds per day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The mSAR is calculated as

$$mSAR = nSAR_{mobiledata} \times outputpower_{mobiledata} + nSAR_{wifi} \times outputpower_{wifi}$$

where

- $nSAR_{data}$, $nSAR_{wifi}$ are the nSAR (normalized specific absorption rates, in W/kg/W) from data use using mobile data or WiFi networks, respectively
- $outputpower_{data}$, $outputpower_{wifi}$ are the output powers (mW) of the device (mobilephone) using mobile data or WiFi networks, respectively

Value

Momentary SAR (mSAR) from mobile data use in mW/kg

See Also

[mpd_pwr_data\(\)](#), [mpd_pwr_wifi\(\)](#), [mpd_sar_data\(\)](#), [mpd_sar_wifi\(\)](#)

Examples

```
mpd_msar(
  tissue      = "brain",
  duration_low = 4860,
  duration_lowmed = 540,
  duration_medhigh = 4860,
  duration_high = 540,
  use_5g      = FALSE,
  wifi_prop_home = 0.5,
  wifi_prop_work = 0.5,
```

```
wifi_prop_travel = 0.5,
urbanicity      = "suburban",
travel_time     = 1800)
```

mpd_pwr_data

*Calculate Mobile Phone Output Power during Mobile Data Use***Description**

Calculates the mobile phone output power during mobile data use.

Usage

```
mpd_pwr_data(
  band,
  duration_low,
  duration_lowmed,
  duration_medhigh,
  duration_high,
  urbanicity,
  travel_time,
  params = load_params()
)
```

Arguments

band	Technology (3g, 4g, or 5g)
duration_low	Duration (in seconds per day) of low output power activities
duration_lowmed	Duration (in seconds per day) of low-medium output power activities
duration_medhigh	Duration (in seconds per day) of medium-high output power activities
duration_high	Duration (in seconds per day) of high output power activities
urbanicity	Urbanicity of home / workplace (rural, suburban, or urban)
travel_time	Time spent commuting in seconds per day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends in the technology used, the location in which the mobile data network (urbanicity, indoors/outdoors/commuting), and the activity the phone is used for.

Value

Output power in mW

Examples

```
mpd_pwr_data(  
  band          = "4g",  
  duration_low  = 4860,  
  duration_lowmed = 540,  
  duration_medhigh = 4860,  
  duration_high = 540,  
  urbanicity    = "rural",  
  travel_time   = 1800)
```

mpd_pwr_wifi	<i>Calculate Mobile Phone Output Power during WiFi Use</i>
--------------	--

Description

Calculates the mobile phone output power during WiFi use

Usage

```
mpd_pwr_wifi(  
  band,  
  duration_low,  
  duration_lowmed,  
  duration_medhigh,  
  duration_high,  
  params = load_params()  
)
```

Arguments

- band WiFi frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz)
- duration_low Duration (in seconds per day) of low output power activities
- duration_lowmed Duration (in seconds per day) of low-medium output power activities
- duration_medhigh Duration (in seconds per day) of medium-high output power activities
- duration_high Duration (in seconds per day) of high output power activities
- params Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends on the frequency band (2.4 GHz or 5.0 GHz) and the type of activity.

Value

Output power in mW

Examples

```
mpd_pwr_wifi(  
  band          = "2",  
  duration_low  = 4860,  
  duration_lowmed = 540,  
  duration_medhigh = 4860,  
  duration_high  = 540)
```

mpd_sar_data	<i>Calculate nSAR during mobile data use on mobile phone</i>
--------------	--

Description

Calculates tissue-specific nSAR from mobile data use on mobile phone.

Usage

```
mpd_sar_data(band, tissue, params = load_params())
```

Arguments

band	Technology ("3g", "4g", or "5g")
tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue and the frequency band.

Value

nSAR in W/kg/W

Examples

```
mpd_sar_data(  
  tissue = "brain",  
  band   = "5g")
```

mpd_sar_wifi	<i>Calculate nSAR during WiFi Use on mobile phone</i>
--------------	---

Description

Calculates tissue-specific nSAR from WiFi use on mobile phone.

Usage

```
mpd_sar_wifi(band, tissue, params = load_params())
```

Arguments

band	Frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz)
tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue and the frequency band.

Value

nSAR in W/kg/W

Examples

```
mpd_sar_wifi(
  tissue = "brain",
  band = "2")
```

other_dose_wrapper	<i>Calculate RF-EMF Dose from Other Sources</i>
--------------------	---

Description

Wrapper function that calculates total dose from additional devices and activities, as used in the GOLIAT consortium study dose calculation.

Usage

```
other_dose_wrapper(
  tissue,
  duration_hotspot,
  duration_smartwatch,
  duration_vr,
  duration_headphones,
  duration_gaming,
  params = load_params()
)
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
duration_hotspot	Duration of time hotspot was active in s/day
duration_smartwatch	Duration of time smartwatch was used in s/day
duration_vr	Duration of time VR headset was used in s/day
duration_headphones	Duration of time Bluetooth headphones were used in s/day
duration_gaming	Duration of gaming with portable console in s/day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from other sources takes into account 5 additional sources:

1. Hotspot (smartphone)
2. Smart watch
3. Virtual reality (VR) headset
4. Bluetooth headphones
5. Portable gaming consoles

The dose is calculated as

$$Dose_{other} = Dose_{hotspot} + Dose_{smartwatch} + Dose_{vr} + Dose_{headphones} + Dose_{gaming}$$

The dose from each source is calculated from the corresponding dose functions.

Value

Tissue-specific RF-EMF dose from other sources in mJ/kg/day

See Also

[hotspot_dose\(\)](#), [smartwatch_dose\(\)](#), [vr_dose\(\)](#), [headphones_dose\(\)](#), [gaming_dose\(\)](#)

Examples

```
other_dose_wrapper(
  tissue      = "brain",
  duration_hotspot = 600,
  duration_smartwatch = 86400,
  duration_vr      = 0,
  duration_headphones = 8280,
  duration_gaming   = 3600)
```

smartwatch_dose	<i>Calculate RF-EMF Dose from Smart Watch</i>
-----------------	---

Description

Calculates the tissue-specific RF-EMF dose from smart-watch use

Usage

```
smartwatch_dose(tissue, duration_smartwatch, params = load_params())
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
duration_smartwatch	duration of time smartwatch was worn in s/day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from smart-watch use is calculated as:

$$Dose_{smartwatch} = Dose_{watch} + Dose_{phone}$$

Where:

- $Dose_{watch}$ is the dose contribution from the smart-watch itself
- $Dose_{phone}$ is the dose contribution from the phone connection to the smart-watch

Value

Tissue-specific RF-EMF dose from smart-watch use in mJ/kg/day

See Also

[smartwatch_watch_dose\(\)](#), [smartwatch_phone_dose\(\)](#)

Examples

```
smartwatch_dose(
  tissue          = "body",
  duration_smartwatch = 6000)
```

smartwatch_phone_dose *Calculate RF-EMF Dose from Smart Watch (phone contribution only)*

Description

Calculates the tissue-specific RF-EMF dose from smart-watch use (phone contribution only).

Usage

```
smartwatch_phone_dose(tissue, duration_smartwatch, params = load_params())
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
duration_smartwatch	duration of time smart-watch was worn in s/day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from smart-watch use (phone contribution) is calculated as:

$$Dose_{phone} = mSAR_{phone} * duration_{phone}$$

Where:

- $mSAR_{phone}$ is the momentary SAR
- $duration_{phone}$ is the use duration

Value

Tissue-specific RF-EMF dose from smart-watch use (phone contribution only) in mJ/kg/day

Examples

```
smartwatch_phone_dose(
  tissue          = "body",
  duration_smartwatch = 6000)
```

smartwatch_watch_dose *Calculate RF-EMF Dose from Smart Watch (watch contribution only)*

Description

Calculates the tissue-specific RF-EMF dose from smart-watch use (watch contribution only).

Usage

```
smartwatch_watch_dose(tissue, duration_smartwatch, params = load_params())
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
duration_smartwatch	duration of time smart-watch was worn in s/day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from smart-watch use (watch contribution) is calculated as:

$$Dose_{watch} = mSAR_{watch} * duration_{watch}$$

Where:

- $mSAR_{watch}$ is the momentary SAR
- $duration_{watch}$ is the use duration

Value

Tissue-specific RF-EMF dose from smart-watch use (watch contribution only) in mJ/kg/day

Examples

```
smartwatch_watch_dose(
  tissue          = "body",
  duration_smartwatch = 6000)
```

tablet_dose

*Calculate RF-EMF Dose from Tablet Use***Description**

Calculates the tissue-specific RF-EMF dose from tablet use.

Usage

```
tablet_dose(
  tissue,
  dur_low,
  dur_lowmed,
  dur_medhigh,
  dur_high,
  params = load_params()
)
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
dur_low	Duration (in seconds per day) of low output power activities on tablet
dur_lowmed	Duration (in seconds per day) of low-medium output power activities on tablet
dur_medhigh	Duration (in seconds per day) of medium-high output power activities on tablet
dur_high	Duration (in seconds per day) of high output power activities on tablet
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The tablet RF-EMF dose is calculated as:

$$Dose_{tablet} = mSAR_{tablet} * duration_{tablet}$$

Where:

- $Dose_{tablet}$ is the dose from tablet use
- $mSAR_{tablet}$ is the momentary SAR value in mJ/kg
- $duration_{tablet}$ is the duration of tablet use

We distinguish between 4 types of activities:

1. Low output power: sending e-mails, browsing the internet, scrolling and chatting on social media, sending text messages
2. Low to medium output power: online gaming, streaming music, sending voice messages
3. Medium to high output power: watching videos, uploading pictures or videos, making video calls
4. High output power: uploading large files

Value

Tissue-specific RF-EMF dose from tablet use in mJ/kg/day

See Also

[tablet_msar\(\)](#)

Examples

```
tablet_dose(  
  tissue      = "brain",  
  dur_low     = 0,  
  dur_lowmed  = 681,  
  dur_medhigh = 0,  
  dur_high    = 0)
```

tablet_msar	<i>Calculate mSAR from Tablet Use</i>
-------------	---------------------------------------

Description

Calculates the tissue-specific momentary SAR (mSAR) from tablet use.

Usage

```
tablet_msar(  
  tissue,  
  dur_low,  
  dur_lowmed,  
  dur_medhigh,  
  dur_high,  
  params = load_params()  
)
```

Arguments

tissue	Tissue for which to calculate mSAR (default: "brain" or "body")
dur_low	Duration (in seconds per day) of low output power activities on tablet
dur_lowmed	Duration (in seconds per day) of low-medium output power activities on tablet
dur_medhigh	Duration (in seconds per day) of medium-high output power activities on tablet
dur_high	Duration (in seconds per day) of high output power activities on tablet
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The mSAR is calculated as:

$$mSAR_{tablet} = nSAR_{tablet} * outputpower_{tablet}$$

Where:

- $mSAR_{tablet}$ is the mSAR from tablet use
- $nSAR_{tablet}$ is the normalised SAR value in mJ/kg
- $outputpower_{tablet}$ is the duration of tablet use

We distinguish between 4 types of activities:

1. Low output power: sending e-mails, browsing the internet, scrolling and chatting on social media, sending text messages
2. Low to medium output power: online gaming, streaming music, sending voice messages
3. Medium to high output power: watching videos, uploading pictures or videos, making video calls
4. High output power: uploading large files

Value

Tissue-specific mSAR from tablet use in mW/kg

See Also

`tablet_pwr()`, `tablet_sar()`

Examples

```
tablet_msar(
  tissue      = "brain",
  dur_low     = 0,
  dur_lowmed  = 1230,
  dur_medhigh = 0,
  dur_high    = 0)
```

tablet_pwr	Calculate Tablet Output Power
------------	-------------------------------

Description

Calculates the tablet output power during use

Usage

```
tablet_pwr(  
  band,  
  dur_low,  
  dur_lowmed,  
  dur_medhigh,  
  dur_high,  
  params = load_params()  
)
```

Arguments

band	WiFi frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz)
dur_low	Duration (in seconds per day) of low output power activities on tablet
dur_lowmed	Duration (in seconds per day) of low-medium output power activities on tablet
dur_medhigh	Duration (in seconds per day) of medium-high output power activities on tablet
dur_high	Duration (in seconds per day) of high output power activities on tablet
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends on the frequency band (2.4 GHz or 5.0 GHz) and the type of activity.

Value

Output power in mW

Examples

```
tablet_pwr(  
  band      = "5",  
  dur_low   = 0,  
  dur_lowmed = 681,  
  dur_medhigh = 0,  
  dur_high  = 0)
```

tablet_sar	<i>Calculate nSAR during Tablet Use</i>
------------	---

Description

Calculates tissue-specific nSAR from tablet use.

Usage

```
tablet_sar(tissue, band, params = load_params())
```

Arguments

tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
band	Frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz)
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue and the frequency band.

Value

nSAR in W/kg/W

Examples

```
tablet_sar(  
  tissue = "brain",  
  band   = "2")
```

total_dose

Calculate Total RF-EMF Dose from All Sources

Description

Calculates the total RF-EMF dose from all available RF-EMF sources for a specific tissue.

Usage

```
total_dose(sample, tissue, params = load_params())
```

Arguments

sample	A list with input values for a single observation. List elements must match the names and data types of the provided example dataset and must not be NA.
tissue	Tissue for which to calculate dose (default: "brain" or "body")
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

A detailed description of the dose calculation will follow.

Value

Total RF-EMF dose in mJ/kg/day for the specified tissue.

vr_dose

*Calculate RF-EMF Dose from Virtual Reality Headset***Description**

Calculates the tissue-specific RF-EMF dose from VR headset

Usage

```
vr_dose(tissue, duration_vr, params = load_params())
```

Arguments

tissue	Tissue for which to calculate dose (default: "brain" or "body")
duration_vr	duration of time VR headset was used in s/day
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from VR headset is calculated as:

$$Dose_{vr} = mSAR_{vr} * duration_{vr}$$

Where:

- $mSAR_{vr}$ is the mSAR of virtual reality headset use
- $Dduration_{vr}$ is the duration of VR headset use in s/day

Value

Tissue-specific RF-EMF dose from VR headset use in mJ/kg/day

Examples

```
vr_dose(
  tissue      = "brain",
  duration_vr = 600)
```

wifi_dose

*Calculate RF-EMF Dose from WiFi Router***Description**

Calculates the tissue-specific RF-EMF dose from WiFi router

Usage

```
wifi_dose(tissue, travel_time, wifi_prop_travel, params = load_params())
```

Arguments

tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
travel_time	Time spent commuting in seconds per day
wifi_prop_travel	Proportion of time connected to WiFi (vs mobile data) while commuting
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The RF-EMF dose from WiFi router is calculated as:

$$Dose_{WiFi} = mSAR_{farfield} * duration_{WiFi}$$

Where:

- $Dose_{WiFi}$ is the dose from WiFi Router
- $mSAR_{WiFi}$ is the momentary SAR value in mJ/kg
- $duration_{WiFi}$ is the duration of time exposed to a WiFi router in seconds/day

Value

Tissue-specific RF-EMF dose from Wifi router in mJ/kg/day

See Also

[wifi_msar\(\)](#)

Examples

```
wifi_dose(
  tissue      = "body",
  travel_time = 1800,
  wifi_prop_travel = 0.5)
```

wifi_msar

*Calculate mSAR of WiFi Router***Description**

Calculates the mSAR of WiFi Router

Usage

```
wifi_msar(tissue, travel_time, wifi_prop_travel, params = load_params())
```

Arguments

tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
travel_time	Time spent commuting in seconds per day
wifi_prop_travel	Proportion of time connected to WiFi (vs mobile data) while commuting
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends on the time spent commuting and the proportion of time a WiFi connection (vs mobile data) is used while commuting.

Value

mSAR in W/kg/W/m**2

See Also

[wifi_pwr\(\)](#), [wifi_sar\(\)](#)

Examples

```
wifi_msar(
  tissue      = "body",
  travel_time = 1800,
  wifi_prop_travel = 0.5)
```

wifi_pwr*Calculate Output Power of WiFi Router*

Description

Calculates the output power of WiFi router

Usage

```
wifi_pwr(band, params = load_params())
```

Arguments

band	Frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz)
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The output power depends on the WiFi frequency band.

Value

Output power in mW/m**2

Examples

```
wifi_pwr(  
band = "2")
```

wifi_sar*Calculate nSAR from WiFi Router*

Description

Calculates tissue-specific nSAR from WiFi Router

Usage

```
wifi_sar(tissue, band, params = load_params())
```

Arguments

tissue	Tissue for which to calculate nSAR (default: "brain" or "body")
band	Frequency band ("2" for 2.4 GHz, "5" for 5.0 GHz)
params	Parameter list (optional). If not specified, calculations use default parameters.

Details

The normalized specific absorption rate (nSAR) depends on the tissue and the WiFi frequency band.

Value

nSAR in W/kg/W/m**2

Examples

```
wifi_sar(  
  tissue = "brain",  
  band   = "5")
```

Index

* datasets

- [example_data](#), [12](#)
- [act_pwr_props](#), [3](#)
- [calculate_emf_doses](#), [3](#)
- [check_boolean_not_na](#), [4](#)
- [check_character_not_na](#), [5](#)
- [check_country](#), [5](#)
- [check_duration](#), [6](#)
- [check_headp_num](#), [6](#)
- [check_numeric_not_na](#), [7](#)
- [check_proportions](#), [7](#)
- [check_tissue](#), [8](#)
- [check_urbanicity](#), [8](#)
- [cordless_dose](#), [9](#)
- [dect_msar](#), [10](#)
- [dect_msar\(\)](#), [9](#)
- [dect_pwr](#), [11](#)
- [dect_pwr\(\)](#), [10](#)
- [dect_sar](#), [11](#)
- [dect_sar\(\)](#), [10](#)
- [example_data](#), [12](#)
- [farfield_dose](#), [13](#)
- [farfield_msar](#), [14](#)
- [farfield_msar\(\)](#), [13](#)
- [farfield_pwr](#), [15](#)
- [farfield_pwr\(\)](#), [14](#)
- [farfield_sar](#), [15](#)
- [farfield_sar\(\)](#), [14](#)
- [fill_missing_variables](#), [16](#)
- [gaming_dose](#), [17](#)
- [gaming_dose\(\)](#), [45](#)
- [headphones_dose](#), [18](#)
- [headphones_dose\(\)](#), [45](#)
- [headphones_earset_dose](#), [19](#)
- [headphones_earset_dose\(\)](#), [18](#)
- [headphones_phone_dose](#), [20](#)
- [headphones_phone_dose\(\)](#), [18](#)
- [hotspot_dose](#), [21](#)

- [hotspot_dose\(\)](#), [45](#)
- [laptop_dose](#), [22](#)
- [laptop_msar](#), [23](#)
- [laptop_msar\(\)](#), [23](#)
- [laptop_pwr](#), [24](#)
- [laptop_pwr\(\)](#), [24](#)
- [laptop_sar](#), [25](#)
- [laptop_sar\(\)](#), [24](#)
- [load_params](#), [26](#)
- [load_params\(\)](#), [8](#), [27](#)
- [load_tissue_params](#), [27](#)
- [load_tissue_params\(\)](#), [27](#)
- [location_props](#), [27](#)
- [mobilecall_dose](#), [28](#)
- [mobiledata_dose](#), [30](#)
- [mpc_bt_earset_msar](#), [31](#)
- [mpc_bt_earset_msar\(\)](#), [32](#)
- [mpc_bt_phone_msar](#), [32](#)
- [mpc_bt_phone_msar\(\)](#), [32](#)
- [mpc_msar](#), [33](#)
- [mpc_msar\(\)](#), [29](#)
- [mpc_pwr_data](#), [34](#)
- [mpc_pwr_data\(\)](#), [34](#)
- [mpc_pwr_native](#), [35](#)
- [mpc_pwr_native\(\)](#), [34](#)
- [mpc_pwr_wifi](#), [36](#)
- [mpc_pwr_wifi\(\)](#), [34](#)
- [mpc_sar_data](#), [36](#)
- [mpc_sar_data\(\)](#), [34](#)
- [mpc_sar_native](#), [37](#)
- [mpc_sar_native\(\)](#), [34](#)
- [mpc_sar_wifi](#), [38](#)
- [mpc_sar_wifi\(\)](#), [34](#)
- [mpd_msar](#), [39](#)
- [mpd_msar\(\)](#), [31](#)
- [mpd_pwr_data](#), [41](#)
- [mpd_pwr_data\(\)](#), [40](#)
- [mpd_pwr_wifi](#), [42](#)
- [mpd_pwr_wifi\(\)](#), [40](#)
- [mpd_sar_data](#), [43](#)
- [mpd_sar_data\(\)](#), [40](#)
- [mpd_sar_wifi](#), [44](#)

`mpd_sar_wifi()`, [40](#)

`other_dose_wrapper`, [44](#)

`smartwatch_dose`, [46](#)

`smartwatch_dose()`, [45](#)

`smartwatch_phone_dose`, [47](#)

`smartwatch_phone_dose()`, [46](#)

`smartwatch_watch_dose`, [48](#)

`smartwatch_watch_dose()`, [46](#)

`tablet_dose`, [49](#)

`tablet_msar`, [50](#)

`tablet_msar()`, [50](#)

`tablet_pwr`, [51](#)

`tablet_pwr()`, [51](#)

`tablet_sar`, [52](#)

`tablet_sar()`, [51](#)

`total_dose`, [53](#)

`vr_dose`, [54](#)

`vr_dose()`, [45](#)

`wifi_dose`, [55](#)

`wifi_msar`, [56](#)

`wifi_msar()`, [55](#)

`wifi_pwr`, [57](#)

`wifi_pwr()`, [56](#)

`wifi_sar`, [57](#)

`wifi_sar()`, [56](#)